# ARMAAN GUHA

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## Why Computer Science + Cyber Security

I am drawn to cybersecurity because of the questions I already chase in my own projects - how do you tell a real signal from an anomaly, how do you protect a system before it fails, how do you make critical infrastructure trustworthy, at their core, security questions? I have been tinkering with computers ever since I was in the 5th grade and I think I have finally found my true passion in Cybersecurity.

## EDUCATION

**Delhi Public School, Noida** - Class 12, Science Stream (Physics, Chemistry, Math, Computer Science, English, AI)

**SAT** - 1500

## PROJECTS

**AcousticLeakNet** - Simulation-trained deep learning system for acoustic water pipe leak detection

- Trained on 100.5M simulated data rows across 8 water distribution networks; reached perfect AUROC/F1 on 3 unseen test networks
- Hardware cost 10–42x cheaper than commercial leak-detection alternatives.
- Co-authored technical research paper: project selected for GENIUS Olympiad 2026 (AI category)

**YEMU** - Malware analysis sandbox for Linux and Windows (GPCSSI)

- Runs samples in disposable QEMU VMs (KVM on Linux, WHPX on Windows) with strace behavioural monitoring, network capture, and static + in-memory YARA scanning
- Explainable 0–100 threat score and verdict; Qt desktop app and CLI, shipped as v0.6.1 with installers and CI

**Log Anomaly Detector** - AI pipeline for detecting abnormal system log behaviour (GPCSSI)

- Collaborative project combining an LSTM sequence model, explainable AI (SHAP), and automated log parsing (Drain3) to flag anomalous event sequences
- Built a web frontend for uploading logs and visualizing detected anomalies

**KrishiMitra** - Digital assistant for Indian farmers

- Interactive dashboard for soil health analysis, AI crop recommendations, weather forecasts, and market price tracking, with multilingual support

**termaan** - Terminal-style personal portfolio website

- Custom command system, tab autocomplete, and history navigation built in vanilla JavaScript

## HONORS & ACHIEVEMENTS

- GENIUS Olympiad 2026 Finalist & Stage 2 Stockholm Junior Water Prize - Artificial Intelligence category, for AcousticLeakNet
- Co-authored an original research paper on acoustic leak detection using a novel 100M+ row simulation dataset
- Active member, science and technology clubs, DPS Noida.
- President of eSpice (2026-27), Joint Secretary of eSpice (Tech Club)(2025-26), Mentoring students in fields like AI/ML, Web Dev, Game Dev and Hardware.
- Participated in 50+ Interschool level competitions (past 3 years)
- Promoted to the Post Graduate Level in Gurugram Police Cyber Security Summer Internship (GPCSSI).

## SKILLS

**Programming:** Python, C/C++, JavaScript, HTML/CSS, SQL

**AI / ML:** Deep learning, PyTorch, TensorFlow, data analysis (pandas), signal processing

**Tools:** Git/GitHub, LaTeX, QEMU

## INTERESTS

Applied AI research, signal processing and anomaly detection, infrastructure, and systems security.